



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Disaster Relief Management System

Semester: Spring 25-26

<i>Group:03</i>		<i>Section: BB</i>
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1. PROJECT PROPOSAL

1.1 Background to the Problem

Bangladesh is known worldwide as highly vulnerable to natural disasters, because of its low elevation in the Ganges-Brahmaputra-Meghna delta and its exposure to marine hazards from the Bay of Bengal. According to the national disaster management bureau's records, the country has had over 200 major disaster events in the past thirty years. These events are of a massive scale and some of the most notable are Cyclone Sidr (2007), Cyclone Aila (2009) and the catastrophic 2022 Sylhet floods that displaced millions of people.

1.2 Selection of Process Model

The software development process model for the Disaster Relief Management System was decided to be the Scrum framework.

Overview of Scrum

Scrum is an Agile iterative and incremental process model that divides the software development into three well-defined phases: the Pre-game phase, the Development (Game) phase and the Post-game phase.

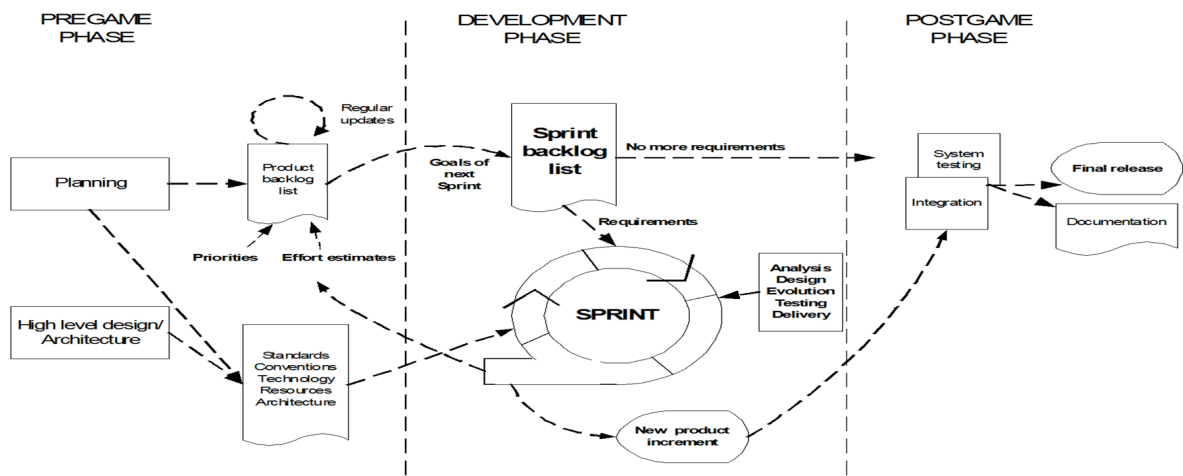


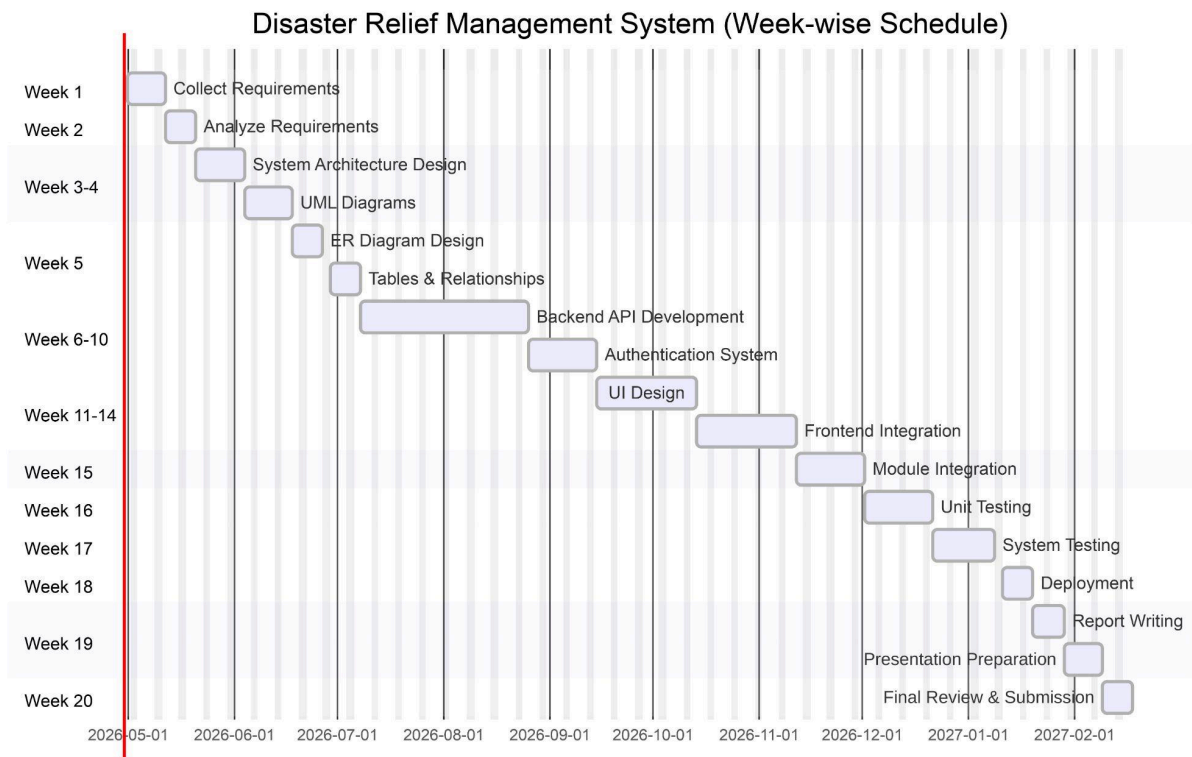
Fig 1: Scrum Process

Development is done in a series of fixed-length cycles called Sprints. The length of a Sprint can be anything between one and four weeks, but not more than one month. At the end of each Sprint a potentially shippable increment of product functionality is created, tested and reviewed. There are three major roles in Scrum, and these include Scrum Team, Scrum Master, and Product Owner. The role of Scrum Master involves ensuring Scrum is implemented in the right manner while the Product Backlog is the responsibility of Product Owner. The Scrum Team performs the planning, construction and delivery in each Sprint.

Reason for Using the SCRUM model

The Disaster Relief Management System was quite complex, the team structure and the time fixed. Thus, Scrum was justified. There are thirty functional requirements divided into four user roles (Admin, Volunteer, Donor and Affected Person) so phased development is necessary. The product backlog list enable effective prioritization and incremental implementation of features. Moreover, the four-member team structure is more appropriate for Scrum roles, which have distinct responsibility distribution and better coordination. Sprints that are time-boxed allow for frequent checks on progress throughout the academic calendar. Advanced features like Face Recognition, Biometric Fraud Detection, and Payment Integration may be continuously tested and improved using the iterative reviews. Regular Scrum meetings also help in communication, synchronization of tasks and resolving issues between interconnected modules. So, Scrum is the best process model for this project.

2. Timeline Chart / Gantt Chart



3. REQUIREMENTS DOCUMENT (PRD)

3.1 Functional Requirements

Functional Requirements			
Item No	Feature	Specifica tion	Screen Definition

FR-01	User Registration [Admin] [Volunteer] [Donor]	Register users with roles	<pre> <form> <input name="full_name"/> <input type="email"/> <input type="password"/> <input type="confirm_password"/ > <select name="role"></select> <button>Register</button> </form> </pre>
FR-02	User Login [Admin] [Volunteer] [Donor]	Secure login system	<pre> <form> <input type="email"/> <input type="password"/> <button>Login</button> <button>Forget Password</button> </form> </pre>
FR-03	Forgot Password [Admin] [Volunteer] [Donor] [Affected Person]	Reset password using OTP	<pre> <form> <input type="email"/> <button>Send OTP</button> <input name="otp"/> <input type="password"/> <button>Reset</button> </form> </pre>
FR-04	Email Verification [Admin] [Volunteer] [Donor] [Affected Person]	Verify user email	<pre> <form> <input name="otp"/> <button>Verify</button> </form> </pre>
FR-05	Profile Manage [Admin] [Volunteer] [Donor] [Affected Person]	update user profile	<pre> <form> <input name="name"/> <input name="phone"/> <input name="address"/> <button>Save</button> </form> </pre>

FR-06	Theme Change [Admin] [Volunteer] [Donor] [Affected Person]	Toggle Light&Dark mode	<button id="themeToggle">Toggle Theme</button>
FR-07	Language Change [Admin] [Volunteer] [Donor] [Affected Person]	Switch language	<select id="lang"> <option>EN</option> <option>BN</option> </select>
FR-08	Admin Dashboard [Admin]	System overview	<div class="dashboard"> <div class="stats"></div> <div class="charts"></div> <div class="map"></div> </div>
FR-09	Manage Relief Camps [Admin]	Add/Edit/ Delete camps	<table> <tr> <td>Camp</td> <td>Location</td> <button>Add</button> <button>Edit</button> </tr> </table>
FR-10	Resource Allocation [Admin]	Allocate resources	<form> <select name="camp"></select> <select name="resource"></select> > <input type="number"/> <button>Allocate</button> </form>
FR-11	Reports & Analytics [Admin]	View system reports	 <canvas id="chart"></canvas> <select name="dateRange"></select>

FR-12	Export Reports [Admin]	Download reports	<button id="exportPDF">Export PDF</button> <button id="exportCSV">Export CSV</button>
FR-13	Manage Users [Admin]	Activate/Deactivate users	<table> <tr> <td>User</td> <button>Activate</button> <button>Deactivate</button> </tr> </table>
FR-14	Volunteer Dashboard [Volunteer]	View assigned tasks	 Task
FR-15	Task Assignment [Admin]	Assign tasks to volunteers	<form> <select name="volunteer"></select> <input name="task"/> <input type="date"/> <button>Assign</button> </form>
FR-16	Task Status Update [Volunteer]	Update task progress	 Task<button>Complete</button>
FR-17	Shift Management [Volunteer]	Manage work shifts	 <select name="shift"></select> <button>Join</button>

FR-18	Performance Tracking [Admin]	Track volunteer performance	 <table> <tr><td>Tasks Done</td></tr> </table>
FR-19	Affected Person Registration [Affected Person]	Register affected users	<form> <input name="name"/> <input name="age"/> <select name="gender"></select> <input name="location"/> <button>Submit</button> </form>
FR-20	Aid Request Submission [Affected Person]	Request aid	<form> <textarea name="needs"></textarea> <input name="members"/> <button>Submit</button> </form>
FR-21	Face Recognition for Aid Registration [Affected Person]	Capture and verify faces of all family members during aid request submission	<form> <input name="members" type="number"/> <div id="faceCapture"> <video id="cameraFeed"></video> > <button id="captureFace">Capture Face</button> <div id="capturedFaces"></div> </div> <button id="verifyFaces">Verify All Faces</button> <button>Submit Request</button>

			</form>								
FR-22	Family Member Verification History [Admin]	Track and display verification history of all family members with timestamps and verification status	class="verification-history"> <table> <tr> <th>Member Name</th> <th>Verification Date</th> <th>Status</th> <th>Confidence Score</th> </tr> <tr> <td colspan="4"> View Details </td> </tr> </table> Download Report	Member Name	Verification Date	Status	Confidence Score	View Details			
Member Name	Verification Date	Status	Confidence Score								
View Details											
FR-23	Biometric Fraud Detection & Alert System [Admin]	Real-time detection of duplicate face registrations across multiple aid requests with automatic admin alerts	class="fraud-detection"> class="alert-panel"> class="fraud-alert"> class="severity">High Risk Duplicate face detected Investigate class="match-details"> original-face duplicate-face Match Confidence: 92% Block Registration								

			<button>Allow (False Positive)</button> </div> </div>
FR-24	Aid Request Tracking [Affected Person]	Track aid status	<table> <tr> <td>Status</td> <button>View</button> </tr> </table>
FR-25	Camp Suggestion [Affected Person]	Suggest nearest camp	<div id="map"></div>
FR-26	Donation Form [Donor]	Donate money or goods	<form> <select name="type"></select> <input name="amount"/> <button>Donate</button> </form>
FR-27	Payment Integration [Donor]	Process payments	<form> <input name="card"/> <button>Pay</button> </form>
FR-28	Donation Tracking [Donor]	Track donations	 <table> <tr><td>Donation</td></tr> </table>
FR-29	Transparency Dashboard [Donor]	Display real-time fund collection, distribution, and utilization	<div class="transparency-dashb oard"> <div class="stats-row"> <div class="stat-card"></div>

		data for donor transparency	
FR-30	Notification System [Admin] [Volunteer] [Donor] [Affected Person]	Send alerts	<div class="notifications"> Alert </div>

3.2 Non-Functional Requirements

	Non-Functional Requirements		
Item No	Category	Requirement	Description
NFR-01	Performance	Page Load Time	All pages must load within acceptable time to ensure smooth UX during disaster events when traffic may spike.
NFR-02	Performance	API Response Time	Backend API endpoints must respond quickly to support real-time dashboard updates.
NFR-03	Performance	Concurrent Users	System must handle simultaneous users during peak disaster response periods.
NFR-04	Performance	Database Query Time	Complex queries (reports, analytics) should execute within acceptable

			thresholds.
NFR-05	Security	Authentication Security	All user accounts must be protected with strong authentication mechanisms.
NFR-06	Security	Data Encryption	Sensitive data (PII, donation info) must be encrypted in transit and at rest.
NFR-07	Security	Role-Based Access Control	Each user role must only access permitted features and data.
NFR-08	Security	Attack Protection	System must be protected against common web attacks.
NFR-09	Security	Payment Security	Payment processing must comply with security standards.
NFR-10	Usability	Responsive Design	System must be usable on mobile, tablet, and desktop devices.
NFR-11	Usability	Multilingual Support	System should support Bengali and English to serve the local disaster-affected population.
NFR-12	Usability	Ease of Use	Interface should be intuitive requiring minimal training.
NFR-13	Reliability	System Uptime	Platform must maintain high availability, especially during active disaster events.
NFR-14	Reliability	Data Backup	All data must be backed up regularly to prevent loss during system failure.
NFR-15	Reliability	Disaster Recovery	System must recover from failures within a defined time window.

NFR-16	Reliability	Error Handling	System must gracefully handle unexpected errors without crashing.
NFR-17	Scalability	Horizontal Scalability	Architecture must support adding servers during high-traffic periods.
NFR-18	Scalability	Database Scalability	Database must handle growing data over multiple disaster events.
NFR-19	Security	Data Privacy	User personal data handled according to privacy regulations.
NFR-20	Security	Audit Logs	All system activities must be tracked and stored for accountability.

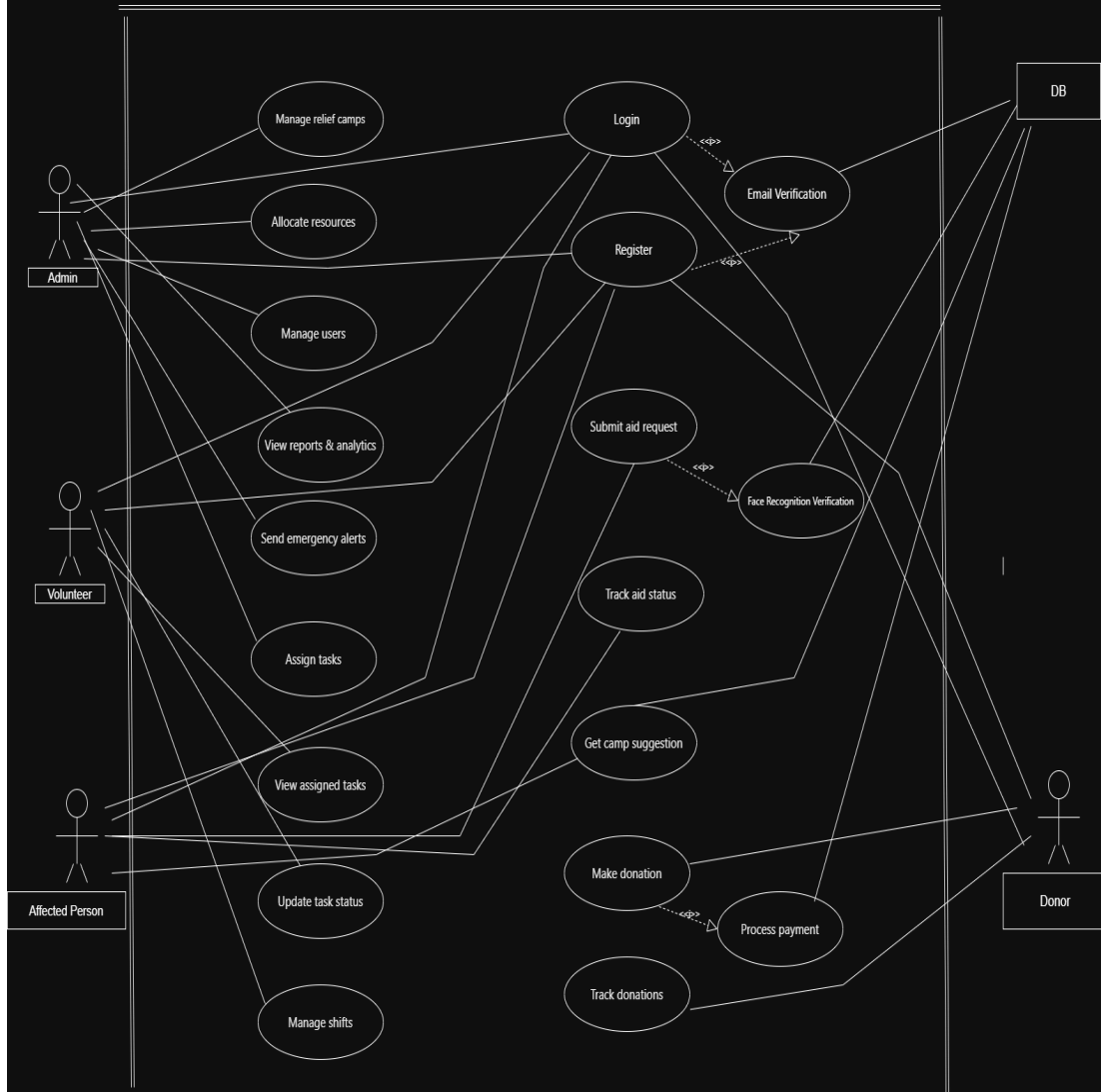
4. SOFTWARE DESIGN

4.1 System Design:

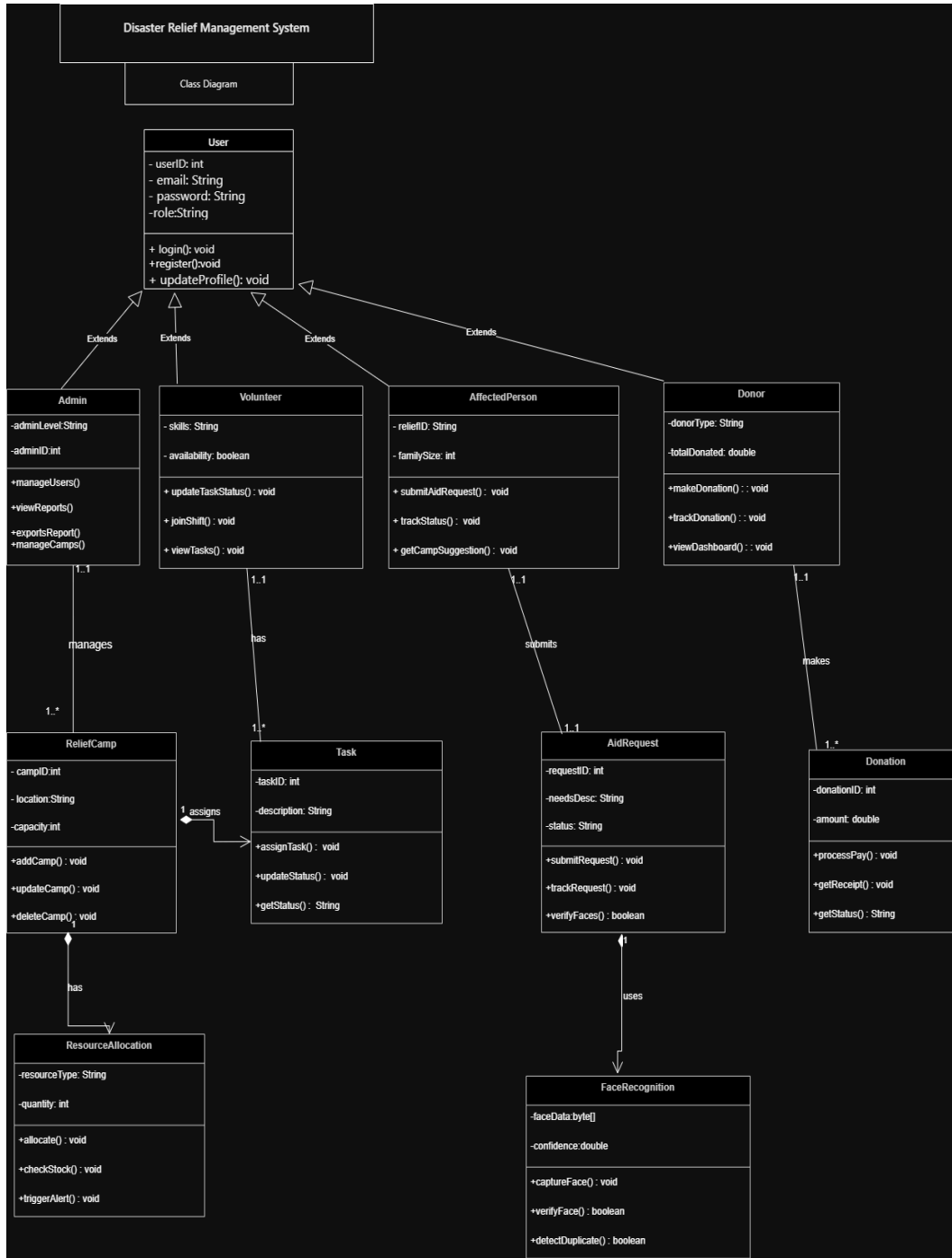
Use Case Diagram

Disaster Relief Management System

Use Case Diagram

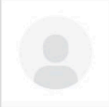


Class Diagram:



4.2 UI Design using Figma

Create Account Full Name Rahim Uddin Email Address you@gmail.com Password Min 8 Character Confirm Password Re-entered password Select Role --Choose Role-- Register Already have an account ? Sign In	Welcome Back Email Address you@gmail.com Password Your password Forget Password? Sign In ----- or ----- Register New Account	Forget Password We'll send an OTP to your email Step-1 - Enter Email Registered Email you@gmail.com Send OTP Step-2 Verify OTP (valid 10 Min) Don't Receive? Resend OTP Step-3 New password New password Min 8 characters Reset password
--	--	--

Verify Your Email OTP Sent to rahim@example.com Verify OTP Don't Receive? Resend OTP OTP expires in 9:23	 Rahim Uddin volunteer Change photo Edit Profile Full Name Rahim Uddin Phone Number +88019367464 Address Dhaka, bangladesh Email (Read-only) rahim@example.com Save Changes	Appearance Customize your display theme Dark Mode Save battery - easier on eyes Dark Theme Light Theme
---	--	--

Language

Choose your preferred language

English

বাংলা

Admin Panel

Dashboard

Live system overview - refreshes every 30s

24Active Camp

1284Affected people

4.2M Funds Raised

912Volunteer

Resources Distribution

Camp Locations

Interactive Map - 24 Camps

Admin

Relief Camps

Manage Active relief camps

+Add Camp

search Camps...

Camp Alpha

Shyghet - Cap: 500

occupancy

Active

EditDelete

Camp Bravo

Cox's Bazar - Cap: 500

occupancy

Near Full

EditDelete

Admin

Resource Allocation

Distribute supplies to active camps

Select Camp

Camp Alpha-shyghet

Resource Type

Food packets

Quantity

Enter Quantity (>0)

Note(Optional)

Add allocation note...

Allocate Resources

Admin

Reports & Analytics

System Performance insights

Date Range

Last 7 days

17Tasks Completed

92%volunteer Rate

volunteer task completion

Admin

Export Reports

Download system data in your format

Report Type

volunteer performance

Date Range

This Month

PDFFormatted report

CSVSpreadsheet data

Export & Download

Admin

Manage Users

ACTIVATE OR Deactivate user accounts

search users...

Filter Role

All Roles

karim Hossain

volunteer. karim@gmail.com

DeactiveView Log

Sumaiya Begum

donor. sumaiya@gmail.com

activateView Log

Volunteer

Hello, karim!

Volunteer, Dhaka Zone

5Active Task

28Completed

Assigned Tasks

Food Distribution -Camp Alpha

Due:Today

Start

Medical Supply Check

Due: Tomorrow

View

Shelter Registration

Due: 15 May

View

Completed tasks archived after 7 days

Admin

Assign Task

Assign tasks to active volunteer

Select Volunteer

karim Hossain- Dhaka

Task Title

e.g Food Distribution

Task Type

Food Distribution

Due Date

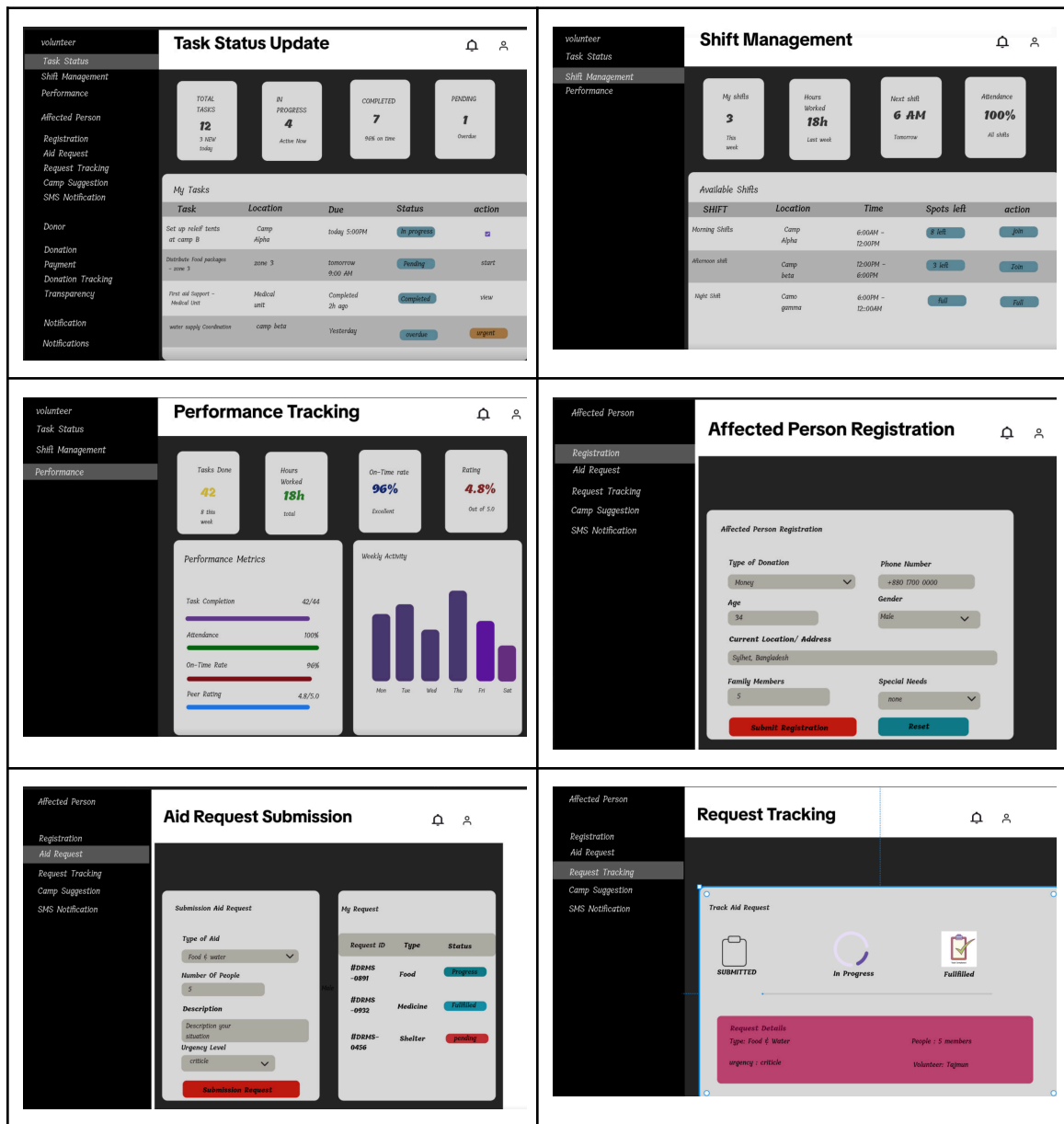
Priority

High

Note(Optional)

Optional task instructions...

Assign Task



volunteer

Task Status

Shift Management

Performance

Performance Tracking

Tasks Done428 this week

Hours Worked18htotal

On-Time rate96%Excellent

Rating4.8%Out of 5.0

Performance Metrics

Task Completion42/44

Attendance100%

On-Time Rate96%

Peer Rating4.8/5.0

Weekly Activity

Bar chart showing activity levels from Mon to Sat.

Affected Person

Registration

Aid Request

Request Tracking

Camp Suggestion

SMS Notification

Affected Person Registration

Type of DonationMoney

Phone Number+880 1700 0000

Age34

GenderMale

Current Location/ AddressDyherf, Bangladesh

Family Members5

Special Needsnone

Submit Registration

Reset

Affected Person

Registration

Aid Request

Request Tracking

Camp Suggestion

SMS Notification

Aid Request Submission

Submission Aid Request

Type of AidFood & water

Number Of People5

DescriptionDisplacement your situation

Urgency Levelcritical

Submission Request

My Request

Request ID	Type	Status
#0045-0091	Food	In progress
#0045-0932	Medicine	Fulfilled
#0045-0456	Shelter	Pending

Affected Person

Registration

Aid Request

Request Tracking

Camp Suggestion

SMS Notification

Request Tracking

Track Aid Request

SUBMITTED

In Progress

Fulfilled

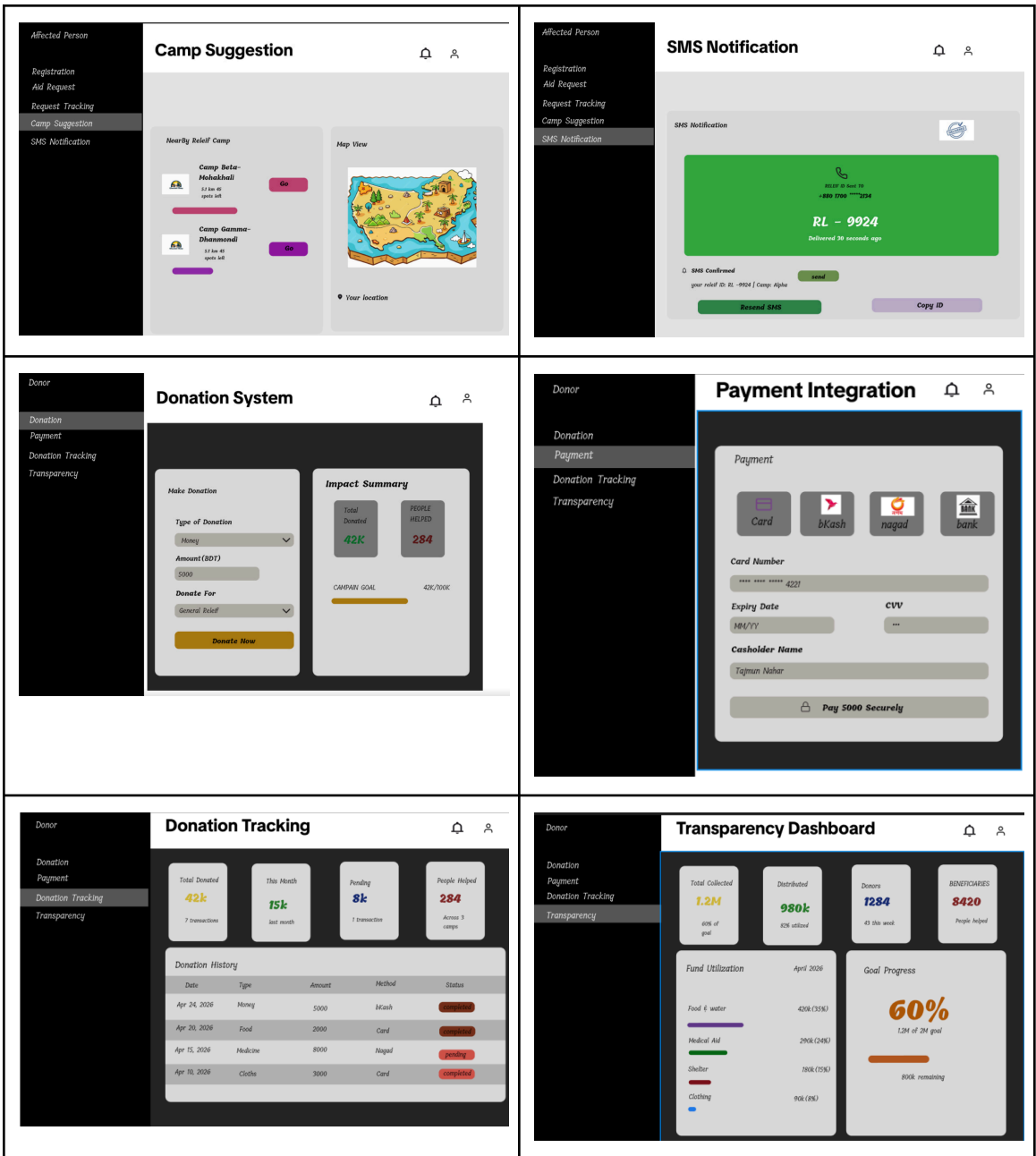
Request Details

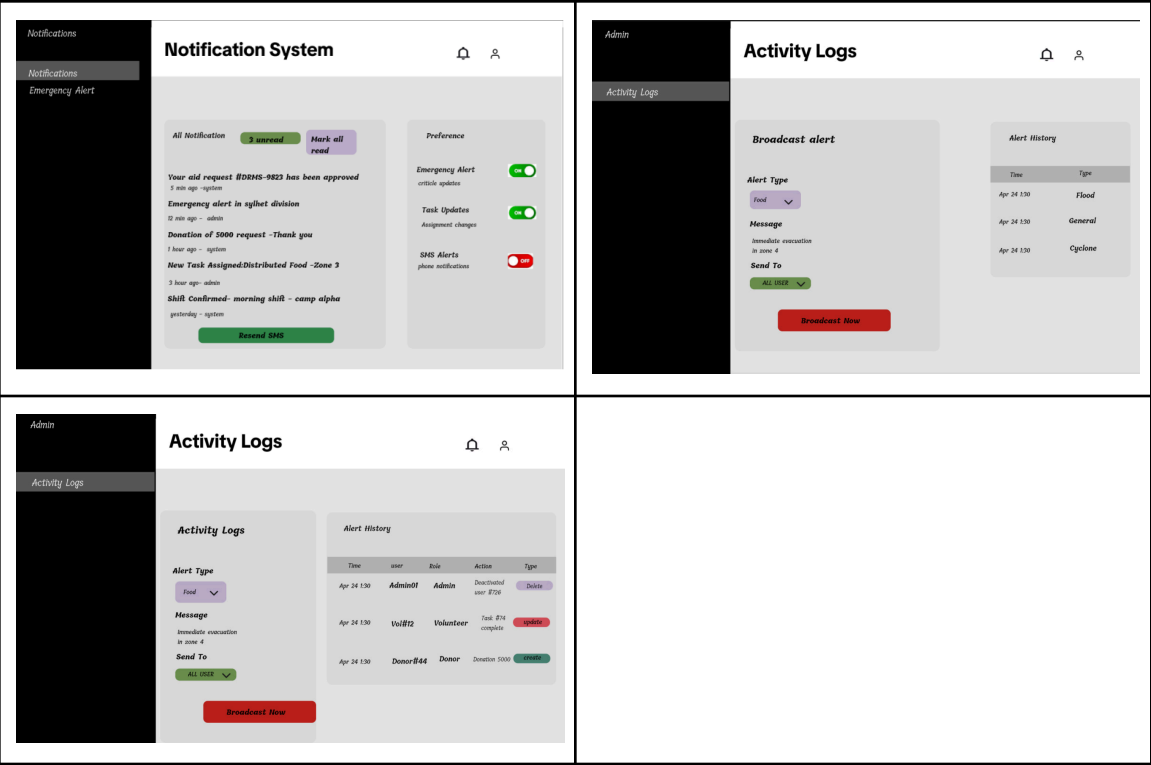
Type: Food & Water

Urgency: critical

People: 5 members

Volunteer: Tajman





5. GIT WORKFLOW

Created a new Repo :

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#)
Required fields are marked with an asterisk (*).

- General**
 - Owner *
shafikulnayem10
 - Repository name *
disaster-relief
✓ disaster-relief is available.
 - Great repository names are short and memorable. How about [vigilant-invention](#)?
 - Description
0 / 350 characters
- Configuration**
 - Choose visibility *
Choose who can see and commit to this repository
Public
 - Add README
READMEs can be used as longer descriptions. [About READMEs](#)
Off
 - Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)
No .gitignore
 - Add license
Licenses explain how others can use your code. [About licenses](#)
No license

Create repository

Initialize files and create branch:

```
C:\Windows\System32\cmd.e x + v
E:\Desktop\disaster-relief>git init
Initialized empty Git repository in E:/Desktop/disaster-relief/.git/
E:\Desktop\disaster-relief>echo "# Disaster Relief Management System" > README.md
E:\Desktop\disaster-relief>echo "# Changelog" > CHANGELOG.md
E:\Desktop\disaster-relief>echo "# Implemented Features" > PROJECT_FEATURES.md
E:\Desktop\disaster-relief>git add README.md CHANGELOG.md PROJECT_FEATURES.md
E:\Desktop\disaster-relief>git commit -m "initial project setup"
[master (root-commit) 552b29a] initial project setup
 3 files changed, 3 insertions(+)
 create mode 100644 CHANGELOG.md
 create mode 100644 PROJECT_FEATURES.md
 create mode 100644 README.md
E:\Desktop\disaster-relief>git branch -M main
E:\Desktop\disaster-relief>git branch dev
E:\Desktop\disaster-relief>git branch stage
E:\Desktop\disaster-relief>git branch
  dev
* main
  stage
```

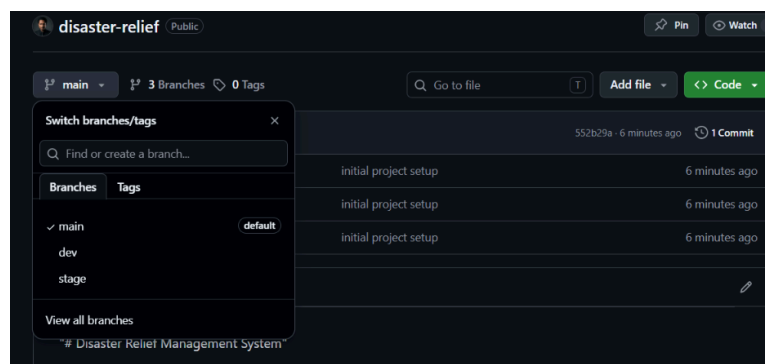
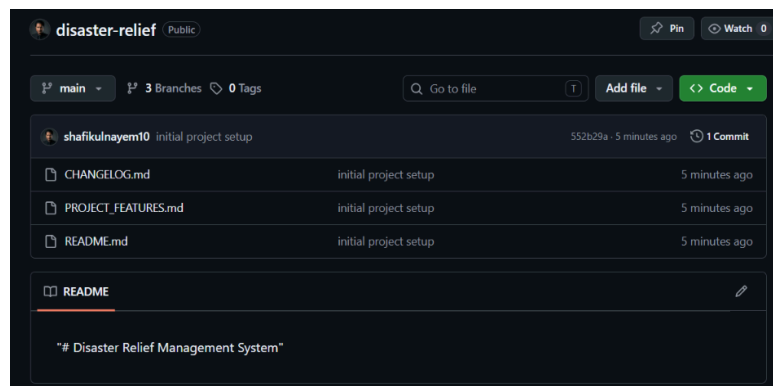
Push files:

```
E:\Desktop\disaster-relief>git remote add origin https://github.com/shafikulnayem10/disaster-relief.git
E:\Desktop\disaster-relief>git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (5/5), 397 bytes | 132.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shafikulnayem10/disaster-relief.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.

E:\Desktop\disaster-relief>git push -u origin dev
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote:   https://github.com/shafikulnayem10/disaster-relief/pull/new/dev
remote:
To https://github.com/shafikulnayem10/disaster-relief.git
 * [new branch]    dev -> dev
branch 'dev' set up to track 'origin/dev'.

E:\Desktop\disaster-relief>git push -u origin stage
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'stage' on GitHub by visiting:
remote:   https://github.com/shafikulnayem10/disaster-relief/pull/new/stage
remote:
To https://github.com/shafikulnayem10/disaster-relief.git
 * [new branch]    stage -> stage
branch 'stage' set up to track 'origin/stage'.

E:\Desktop\disaster-relief>
```



Create Feature Branch:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User\OneDrive\Desktop>git clone https://github.com/shafikulnayem10/disaster-relief.git
Cloning into 'disaster-relief'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.

C:\Users\User\OneDrive\Desktop>cd disaster-relief
C:\Users\User\OneDrive\Desktop\disaster-relief>git checkout dev
branch 'dev' set up to track 'origin/dev'.
Switched to a new branch 'dev'

C:\Users\User\OneDrive\Desktop\disaster-relief>git pull origin dev
From https://github.com/shafikulnayem10/disaster-relief
* branch dev -> FETCH_HEAD
Already up to date.

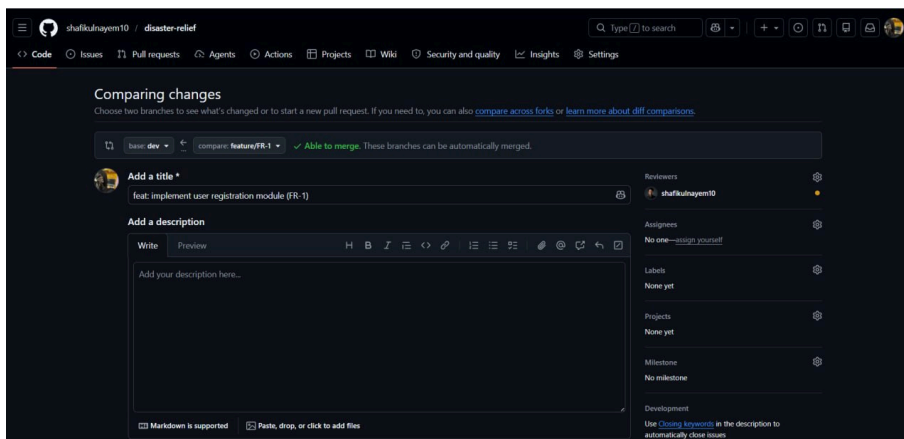
C:\Users\User\OneDrive\Desktop\disaster-relief>git checkout -b feature/FR-1
Switched to a new branch 'feature/FR-1'

C:\Users\User\OneDrive\Desktop\disaster-relief>echo "## FR-1: User Registration Module" >> PROJECT_FEATURES.md
C:\Users\User\OneDrive\Desktop\disaster-relief>echo "Implemented registration with role selection and email verification." >> PROJECT_FEATURES.md
C:\Users\User\OneDrive\Desktop\disaster-relief>echo "***Status: Implemented***" >> PROJECT_FEATURES.md
C:\Users\User\OneDrive\Desktop\disaster-relief>git add PROJECT_FEATURES.md
C:\Users\User\OneDrive\Desktop\disaster-relief>git commit -m "feat: implement user registration module (FR-1)"
[feature/FR-1 08c28f2] feat: implement user registration module (FR-1)
1 file changed, 3 insertions(+)
C:\Users\User\OneDrive\Desktop\disaster-relief>git push -u origin feature/FR-1
remote: Permission to shafikulnayem10/disaster-relief:git denied to shafikulnayem10.
fatal: unable to access 'https://github.com/shafikulnayem10/disaster-relief.git/': The requested URL returned error: 403

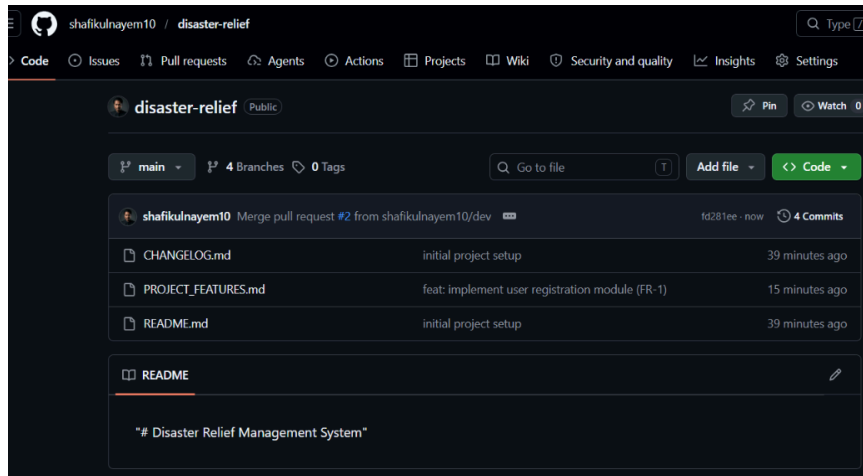
C:\Users\User\OneDrive\Desktop\disaster-relief>git push -u origin feature/FR-1
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 407 bytes | 407.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Create a pull request for 'feature/FR-1' on GitHub by visiting:
remote: https://github.com/shafikulnayem10/disaster-relief/pull/new/feature/FR-1
remote: To https://github.com/shafikulnayem10/disaster-relief.git
     [new branch]   feature/FR-1 -> feature/FR-1
branch 'feature/FR-1' set up to track 'origin/feature/FR-1'.

C:\Users\User\OneDrive\Desktop\disaster-relief>
```

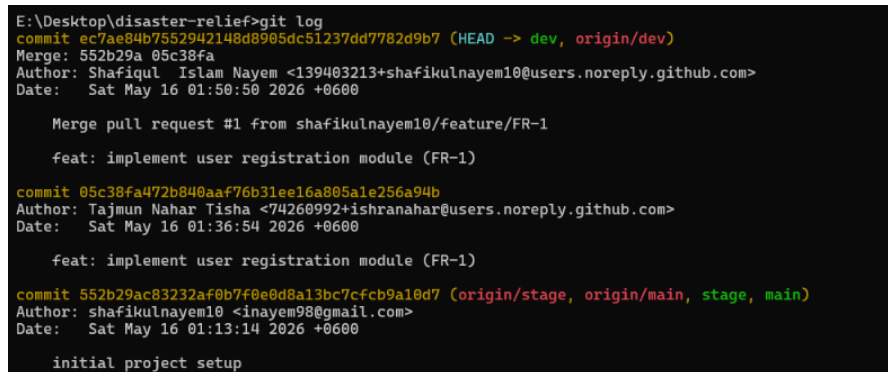
PR Create:



Merge PR:



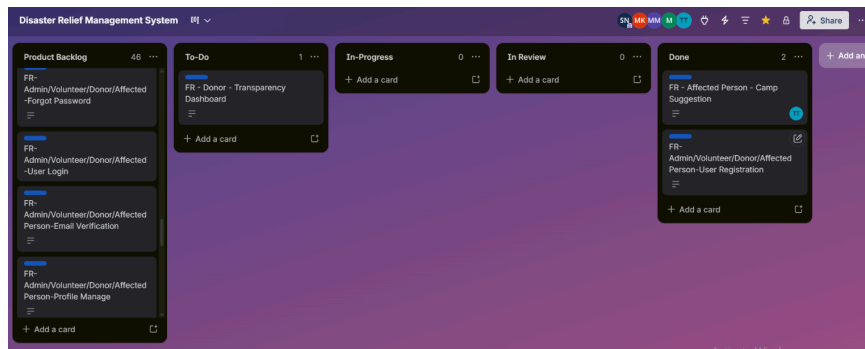
Commit History:



Github Repo Link:

<https://github.com/shafikulnayem10/disaster-relief>

Trello board:



6. SOFTWARE TESTING

6.1 Identified Testing Methods

The following testing methods have been identified in the testing phase of the Disaster Relief Management System.

1. Unit Test

Before combining with other components, the individual modules are tested separately through unit testing. The individual modules, including the User Registration module, the User Login module, the Forgot Password module, the Email Verification module, the Profile Management module, the Theme Change module, and the Language Change module, will be tested separately before being combined into the system.

2. Integration Testing

The aim of integration testing is to assess if the different modules work correctly when integrated. Each module needs to be successful for several features to work in this system. For

example, the Task Status Update module depends on the Task Assignment module and the Face Recognition module requires the Affected Person Registration to be completed first. To ensure that these modules function properly as a unit, integration testing is carried out.

3. System testing

The testing of the entire and fully integrated system to check if it meets the specified requirements. It will be used to validate end-to-end workflows including the full aid request lifecycle, notification delivery to all user roles, the Transparency Dashboard showing real-time fund data, and the Biometric Fraud Detection alert system working in conjunction with the aid registration workflow. The aim is to make sure that Disaster Relief Management System is working properly as a whole system.

6.2 Test Case

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem
Test Case ID: TC_01	Test Designed date: 15/05/2025
Test Priority (Low, Medium, High): High	Test Executed by: Shafiqul Islam Nayem
Module Name: User Registration (FR-01)	Test Execution date:
Test Title: Verify successful registration with valid data for Volunteer role	
Description: Test the user registration page with all valid inputs for Volunteer role	
Precondition: User is on the registration page and not logged in	

Dependencies: Email service must be active				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Registration page 2. Enter Full Name 3. Enter valid Email 4. Enter Password 5. Confirm Password 6. Select Role as Volunteer 7. Click Register button	Name: Nayem Ahmed Email: nayem@mail.com Password: Test@1234 Confirm: Test@1234 Role: Volunteer	User registered successfully and redirected to email verification page		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem			
Test Case ID: TC_02	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): High	Test Executed by: Shafiqul Islam Nayem			
Module Name: User Login (FR-02)	Test Execution date:			
Test Title: Verify login with valid credentials for Admin				
Description: Test the login page with correct email and password for Admin role				
Precondition: Admin account already registered and email verified				
Dependencies: Registration module must be working				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Go to Login page 2. Enter valid Email 3. Enter valid Password 4. Click Login button	Email: admin@drms.com Password: Admin@5678	Admin logged in successfully and redirected to Admin Dashboard		
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Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem			
Test Case ID: TC_03	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): Medium	Test Executed by: Shafiqul Islam Nayem			
Module Name: Forgot Password (FR-03)	Test Execution date:			
Test Title: Verify password reset using OTP with valid registered email				
Description: Test the forgot password flow using OTP sent to registered email				
Precondition: User account exists with a valid registered email				
Dependencies: Email OTP service must be active				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Login page 2. Click Forget Password 3. Enter registered email 4. Click Send OTP 5. Enter received OTP 6. Enter new password 7. Click Reset	Email: donor@mail.com OTP: 482910 New Password: NewPass@10	Password reset successfully; user can login with new password		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem			
Test Case ID: TC_04	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): High	Test Executed by: Tajmun Nahar Tisha			
Module Name: Email Verification (FR-04)	Test Execution date:			
Test Title: Verify email verification with valid OTP after registration				
Description: Test email verification step using the OTP received after user registration				
Precondition: User has just registered and OTP has been sent to email				
Dependencies: Registration module and email service must be working				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Complete Registration 2. Open email inbox 3. Copy the OTP 4. Go to Email Verification page 5. Enter OTP 6. Click Verify button	OTP: 739201	Email verified successfully; user account activated and redirected to login page		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem
Test Case ID: TC_05	Test Designed date: 15/05/2025
Test Priority (Low, Medium, High): Medium	Test Executed by: Tajmun Nahar Tisha

Module Name: Profile Management (FR-05)	Test Execution date:			
Test Title: Verify donor can update profile information successfully				
Description: Test that a logged-in Donor can update their name, phone, and address				
Precondition: Donor is logged in and on the Profile page				
Dependencies: Login module must be working				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Donor 2. Go to Profile page 3. Update Name field 4. Update Phone field 5. Update Address field 6. Click Save button	Name: Tajmun Tisha Phone: 01742365218 Address: Dhaka, Bangladesh	Profile updated successfully with a confirmation message shown		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem
Test Case ID: TC_06	Test Designed date: 15/05/2025
Test Priority (Low, Medium, High): Low	Test Executed by: Tajmun Nahar Tisha
Module Name: Theme Change (FR-06)	Test Execution date:
Test Title: Verify toggle between Light and Dark mode works correctly	
Description: Test the theme toggle feature switches UI between light and dark mode	
Precondition: User is logged in; default theme is Light mode	

Dependencies: User must be logged in				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as any user 2. Locate the Theme Toggle button 3. Click Toggle Theme button 4. Observe UI theme change 5. Click Toggle Theme button again	Initial Mode: Light After 1st Click: Dark After 2nd Click: Light	UI switches to Dark mode on first click and back to Light mode on second click		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem			
Test Case ID: TC_07	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): Low	Test Executed by: MD.Mahedi Hasan			
Module Name: Language Change (FR-07)	Test Execution date:			
Test Title: Verify system language switches from English to Bengali				
Description: Test the language dropdown to switch system language to Bengali (BN)				
Precondition: User is logged in; current language is English (EN)				
Dependencies: Bengali language pack must be loaded				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as any user 2. Locate the Language selector dropdown 3. Click the dropdown	From: EN (English) To: BN (Bengali)	All UI labels and menus switch to Bengali language successfully		

4. Select BN (Bengali) 5. Observe the page content language				
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Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem			
Test Case ID: TC_08	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): High	Test Executed by: MD.Mahedi Hasan			
Module Name: User Registration (FR-01)	Test Execution date:			
Test Title: Verify registration fails when password and confirm password do not match				
Description: Test that the system prevents registration when passwords are mismatched				
Precondition: User is on the Registration page				
Dependencies: None				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Registration page 2. Enter Full Name 3. Enter valid Email 4. Enter Password 5. Enter different Confirm Password 6. Click Register button	Name: Mahedi Hasan Email: mahedi@mail.com Password: Pass@1234 Confirm: Pass@9999	Error message shown: 'Passwords do not match'. Registration not submitted.		

Project Name: Disaster Relief Management System	Test Designed by:Shafiqul Islam Nayem
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Test Case ID: TC_09	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): High	Test Executed by: Mohsina Akter Lova			
Module Name: User Login (FR-02)	Test Execution date:			
Test Title: Verify login fails with incorrect password				
Description: Test that the system rejects a login attempt with a wrong password				
Precondition: User account exists with registered email				
Dependencies: Registration module must be working				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Login page 2. Enter valid registered Email 3. Enter incorrect Password 4. Click Login button	Email: volunteer@gmail.com Password: WrongPass@10	Error message shown: 'Invalid email or password'. The user is not signed in.		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem
Test Case ID: TC_10	Test Designed date: 15/05/2025
Test Priority (Low, Medium, High): Medium	Test Executed by: Shafiqul Islam Nayem
Module Name: Profile Management (FR-05)	Test Execution date:
Test Title: Verify profile update fails when required Name field is left empty	

Description: Test that profile update is blocked if the Name field is cleared and saved				
Precondition: User is logged in and on the Profile Management page				
Dependencies: Login module must be working				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Volunteer 2. Go to Profile page 3. Clear the Name field 4. Update Phone number 5. Click Save button	Name: (empty) Phone: 01833445566	Validation error shown: 'Name is required'. Profile not saved.		

Project Name: Disaster Relief Management System	Test Designed by: Tajmun Nahar Tisha
Test Case ID: TC_11	Test Designed date: 15/05/2025
Test Priority (Low, Medium, High): High	Test Executed by: Shafiqul Islam Nayem
Module Name: Task Assignment (FR-15)	Test Execution date:
Test Title: Verify admin can assign a task to a volunteer with a deadline	
Description: Test that admin can successfully assign a new task to a volunteer from the Task Assignment form	
Precondition: Admin is logged in; at least one Volunteer account exists in the system	
Dependencies: Volunteer registration must be completed	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Admin 2. Navigate to Task Assignment section 3. Select a Volunteer from the dropdown 4. Enter Task description 5. Select a deadline date 6. Click Assign button	Volunteer: Nayem Ahmed Task: Distribute food at Camp-3 Deadline: 20/05/2025	Task assigned successfully; volunteer receives a notification about the new task		

Project Name: Disaster Relief Management System	Test Designed by: Tajmun Nahar Tisha			
Test Case ID: TC_12	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): Medium	Test Executed by: Tajmun Nahar Tisha			
Module Name: Task Status Update (FR-16)	Test Execution date:			
Test Title: Verify volunteer can mark an assigned task as completed				
Description: Test that a volunteer can update the status of an assigned task to complete				
Precondition: Volunteer is logged in; at least one task has been assigned to the volunteer				
Dependencies: FR-15 Task Assignment must be working				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Volunteer 2. Go to Volunteer Dashboard 3. View the assigned task list	Task: Distribute food at Camp-3 Action: Click Complete	Task status updated to Completed; admin dashboard reflects the updated status		

4. Locate the assigned task 5. Click the Complete button next to the task				
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Project Name: Disaster Relief Management System	Test Designed by: Tajmun Nahar Tisha			
Test Case ID: TC_13	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): Medium	Test Executed by: MD.Mahedi Hasan			
Module Name: Shift Management (FR-17)	Test Execution date:			
Test Title: Verify volunteer can join an available shift successfully				
Description: Test that a volunteer can select and join an available work shift				
Precondition: Volunteer is logged in; at least one shift is available in the system				
Dependencies: Admin must have created shift slots beforehand				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Volunteer 2. Navigate to Shift Management section 3. View available shifts 4. Select a shift from the dropdown 5. Click Join button	Shift: Morning (6AM - 12PM) Date: 18/05/2025	Volunteer successfully enrolled in the selected shift; confirmation message displayed		

Project Name: Disaster Relief Management System	Test Designed by: Tajmun Nahar Tisha
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Test Case ID: TC_14	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): High	Test Executed by: Mohsina Akter Lova			
Module Name: Affected Person Registration (FR-19)	Test Execution date:			
Test Title: Verify affected person can register with valid personal information				
Description: Test that an affected person can submit their registration form successfully				
Precondition: Affected person is on the registration page				
Dependencies: System must be accessible without prior login for affected persons				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Affected Person Registration page 2. Enter Name 3. Enter Age 4. Select Gender 5. Enter Location 6. Click Submit button	Name: Rina Begum Age: 35 Gender: Female Location: Sylhet Sadar	Registration submitted successfully; confirmation message shown and data saved in the system		

Project Name: Disaster Relief Management System	Test Designed by: Tajmun Nahar Tisha
Test Case ID: TC_15	Test Designed date: 15/05/2025
Test Priority (Low, Medium, High): High	Test Executed by: Shafiqul Islam Nayem
Module Name: Face Recognition for Aid	Test Execution date:

Registration (FR-21)				
Test Title: Verify face capture and verification works for all family members during aid request				
Description: Test that the face recognition module successfully captures and verifies faces of all family members				
Precondition: Affected person is logged in and on the Aid Request page; camera access is enabled				
Dependencies: FR-19 registration must be completed; face recognition service must be active				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Affected Person 2. Go to Aid Request Submission page 3. Enter number of family members as 3 4. Click Capture Face for each member 5. Click Verify All Faces button 6. Click Submit Request button	Members: 3 Captured: 3 individual face photos	All 3 faces verified successfully; aid request submitted with verification status shown		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem
Test Case ID: TC_16	Test Designed date: 15/05/2025
Test Priority (Low, Medium, High): High	Test Executed by: Tajmun Nahar Tisha

Module Name: Family Member Verification History (FR-22)		Test Execution date:		
Test Title: Verify admin can view verification history with timestamps for all family members				
Description: Test that the admin can access the verification history table with member details and status				
Precondition: Admin is logged in; at least one aid request with face verification has been submitted				
Dependencies: FR-21 Face Recognition must have processed at least one request				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Admin 2. Navigate to Verification History section 3. Select a submitted aid request 4. View the verification history table 5. Check member names, timestamps, status, and confidence score	Aid Request ID: REQ-2025-001 Member: Taslima Akther	Table displays member name, verification date, status, and confidence score for all members		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem
Test Case ID: TC_17	Test Designed date: 15/05/2025
Test Priority (Low, Medium, High): High	Test Executed by: MD.Mahedi Hasan
Module Name: Biometric Fraud Detection (FR-23)	Test Execution date:

Test Title: Verify system detects duplicate face and sends real-time alert to admin				
Description: Test that submitting the same face in two different aid requests triggers a fraud alert				
Precondition: Admin is logged in; one aid request with face data already exists				
Dependencies: FR-21 Face Recognition must be functional; alert panel must be active				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Submit aid request using an already-registered face 2. System processes face recognition 3. Admin opens the Fraud Detection alert panel 4. View the duplicate face match details 5. Click Investigate button	Duplicate face: same person registered twice Match Confidence: 92%	High Risk fraud alert appears with original and duplicate face images and match confidence score		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem
Test Case ID: TC_18	Test Designed date: 15/05/2025
Test Priority (Low, Medium, High): High	Test Executed by: Mohsina Akter Lova
Module Name: Donation Form (FR-26)	Test Execution date:
Test Title: Verify donor can submit a monetary donation with valid amount	
Description: Test the donation form by submitting a valid money donation	

Precondition: Donor is logged in and on the Donation page				
Dependencies: Payment integration module must be active				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Donor 2. Go to Donation Form 3. Select donation type as Money 4. Enter donation amount 5. Click Donate button	Type: Money Amount: 5000 BDT	Donation submitted successfully; donor redirected to payment processing page		

Project Name: Disaster Relief Management System	Test Designed by: Shafiqul Islam Nayem			
Test Case ID: TC_19	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): Medium	Test Executed by: Shafiqul Islam Nayem			
Module Name: Transparency Dashboard (FR-29)	Test Execution date:			
Test Title: Verify donor can view real-time fund collection and distribution data				
Description: Test that the Transparency Dashboard shows up-to-date donation statistics				
Precondition: Donor is logged in; at least one donation has been processed				
Dependencies: Donation module must be active; admin must have updated fund allocation				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Login as Donor 2. Navigate to Transparency Dashboard 3. View fund collection stats 4. View distribution and utilization stat cards 5. Verify data is current	Total Collected: 1,50,000 BDT Distributed: 80,000 BDT	Dashboard shows real-time stats for total fund collected, distributed, and remaining balance		
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Project Name: Disaster Relief Management System	Test Designed by: Shafiquel Islam Nayem			
Test Case ID: TC_20	Test Designed date: 15/05/2025			
Test Priority (Low, Medium, High): Medium	Test Executed by: Tajmun Nahar Tisha			
Module Name: Notification System (FR-30)	Test Execution date:			
Test Title: Verify admin can send alert notification to all users successfully				
Description: Test that the notification system delivers alerts to all user roles				
Precondition: Admin is logged in; at least one user of each role exists in the system				
Dependencies: All user roles must have registered accounts				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Admin 2. Go to Notification section 3. Compose a new alert message 4. Select recipients as All Users 5. Click Send button	Message: 'Cyclone alert: Report to nearest camp' Recipients: All Users	Notification delivered and visible in all users' notification panels with correct message		

6. Login as Volunteer and check notification panel				
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7. CONCLUSION

The Disaster Relief Management System is an efficient and purposeful software that aims to tackle one of the major humanitarian problems that occur regularly in Bangladesh by using contemporary software engineering approaches. Throughout the entire implementation of the system, the team used the Scrum process model that allowed for systematic sprint planning and role assignment throughout the course of all stages of this project. All stages ranging from requirements gathering and proposal preparation to the design, workflow on Git and software testing were done efficiently and with awareness of the actual value that the system will provide to users and those affected by disaster.

The system includes thirty functional requirements and covers four different types of users: Admin, Volunteer, Donor, and Affected Person with advanced functionalities including: Face Recognition aid registration, Biometric Fraud Detection, payment integration and real time Transparency Dashboard. All of these requirements and functionalities were created in order to add responsibility, efficiency and fair aid distribution in countries affected by disasters on a regular basis. Not only did this project help improve the technical skills of the group in aspects like software systems design, database creation, and version control through Git, but it also taught them about the role thoughtfully designed software plays in helping marginalized groups during times of need. The team feels that with more work done on this software, there is potential for this to be a powerful tool for disaster relief in Bangladesh.